YEAR 10 PHYSICS, AQA GCSE, LESSON 7 OF 8

PARTICLE MODEL OF MATTER

LESSON 07

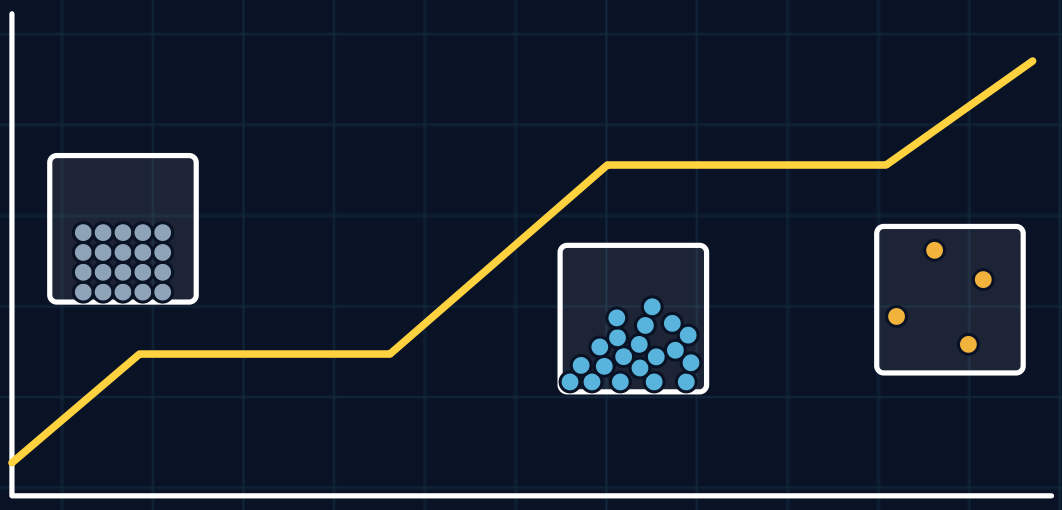
SPECIFIC HEAT CAPACITY

On a sunny beach the sand burns your feet but the sea is still cold. Both have had the same sunshine. Why?

IN THIS LESSON YOU WILL

- state the three things a temperature rise depends on
- state what specific heat capacity means
- calculate with $\Delta E = m\,c\,\Delta\theta$
- decide whether a problem needs specific heat capacity or specific latent heat

ANSWER EDITION FOR TEACHERS AND SELF-MARKING



NAME _____ CLASS _____

START HERE

HOW THIS BOOKLET WORKS

This is the booklet for **Lesson 7** of the Particle Model of Matter unit. It builds on the lessons before it; the “Do now” questions and the quick answers at the back help you check what you remember. Keep the booklet: it is your classwork book *and* your revision guide.

DO NOW
Quick questions from earlier lessons. Remembering again is what makes it stick.

LEARN IT
The explanation, diagrams and worked examples. Read this again when you revise.

TRY IT
Practice that builds up: first with support, then on your own, then a challenge.

PRACTICAL
Record your measurements and work out results from them.

EXAM CORNER
AQA-style questions with a tip on how the marks are given.

REVISE IT
Summary, cover-and-answer questions and an “I can” check to colour in.

! **Two kinds of question to know.** When a sentence has a gap, the choices are printed straight after it, like this: The particles in a solid are close together **(close together / far apart)**. When you see a box of choices, **draw a ring around** the correct one, as AQA exam papers ask. The unit of mass is **the newton** | **the kilogram** | **the joule** .

THE ROUTE THROUGH THE UNIT (THIS LESSON IS HIGHLIGHTED)

01 Solids, liquids and gases

02 Gas particles and pressure

03 Density

04 Measuring density (required practical)

05 Internal energy and heating curves

06 Specific latent heat

07 Specific heat capacity
What a temperature rise depends on, $\Delta E = m\,c\,\Delta\theta$, rearranging, heat capacity or latent heat? (slides 35–39)

08 Measuring specific heat capacity (required practical)

The **equation toolkit** near the back has every equation you have met so far, with formula triangles to help you rearrange. Teaching deck: **panphy.app/gcsephy/decks/particle-model**

Lesson

07 SPECIFIC HEAT CAPACITY

AQA Physics 4.3.2.2, 4.1.1.3; Combined Science 6.3.2.2, 6.1.1.3

BIG QUESTION **On a sunny beach the sand burns your feet but the sea is still cold. Both have had the same sunshine. Why?**

BY THE END YOU CAN

- state the three things a temperature rise depends on
- state what specific heat capacity means
- calculate with $\Delta E = m\,c\,\Delta\theta$
- decide whether a problem needs specific heat capacity or specific latent heat

specific heat capacity **temperature change (Δθ)** **thermal energy** **joule per kilogram per degree Celsius (J/kg °C)**

YOU WILL NEED

Per pair: calculator. Demonstration: two beakers on identical hotplates, one with 200 g of water and one with 200 g of cooking oil, each with a thermometer.

Teaching deck, slides 35–39
panphy.app/gcsephy/decks/particle-model/#35

REMEMBER

DO NOW

From earlier lessons. No notes!

1
Write the equation for the energy needed for a change of state.

$E = m\,L$

2
Which energy of the particles increases when the temperature of a substance rises?

Kinetic energy.

3
Water is heated from 18 °C to 63 °C. Calculate the temperature change.

63 − 18 = 45 °C

EXPLAIN

HOW HARD IS IT TO HEAT?

Heat a substance without changing its state and its temperature rises. How much it rises depends on three things:

- the **energy** transferred to it: more energy, bigger rise

- its **mass**: more mass, smaller rise
- the **material** it is made of: the same energy warms 1 kg of copper far more than 1 kg of water

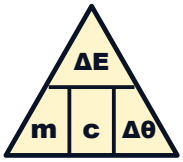
> ! The **specific heat capacity** of a substance is the amount of energy required to raise the temperature of **one kilogram** of the substance by **one degree Celsius**.

$$\Delta E = m\,c\,\Delta\theta$$

change in thermal energy = mass × specific heat capacity × temperature change

ΔE in J, m in kg, c in J/kg °C, $\Delta\theta$ in °C

Rearranged: $c = \dfrac{\Delta E}{m\,\Delta\theta}$ and $\Delta\theta = \dfrac{\Delta E}{m\,c}$. Δ (delta) means “change in”; θ (theta) is temperature.



MATERIAL	SPECIFIC HEAT CAPACITY IN J/KG °C
water	4200
cooking oil	2000
aluminium	900
iron	450
copper	390
lead	130

Water has a very high specific heat capacity: it takes a lot of energy to warm it up, and it gives out a lot as it cools. That is why the sea stays cool on a hot day, and why water is used in radiators and to cool car engines.

WORKED EXAMPLE

Calculate the energy needed to heat 2.0 kg of water from 20 °C to 100 °C. Specific heat capacity of water = 4200 J/kg °C

KNOW	m = 2.0 kg, c = 4200 J/kg °C, $\Delta\theta = 100 - 20 = 80$ °C
EQUATION	$\Delta E = m\,c\,\Delta\theta$
SUBSTITUTE	$\Delta E = 2.0 \times 4200 \times 80$
ANSWER	$\Delta E = 672\,000$ J

YOUR TURN

Calculate the energy needed to raise the temperature of a 0.50 kg aluminium pan by 30 °C. Specific heat capacity of aluminium = 900 J/kg °C

KNOW	**m = 0.50 kg, c = 900 J/kg °C, $\Delta\theta = 30$ °C**
EQUATION	$\Delta E = m\,c\,\Delta\theta$
SUBSTITUTE	$\Delta E = 0.50 \times 900 \times 30$
ANSWER	$\Delta E = 13\,500$ J

FINDING THE TEMPERATURE CHANGE OR THE SPECIFIC HEAT CAPACITY

WORKED EXAMPLE

7800 J is transferred to a 0.50 kg block of copper. Calculate its temperature rise. Specific heat capacity of copper = 390 J/kg °C

KNOW	$\Delta E = 7800$ J, m = 0.50 kg, c = 390 J/kg °C
REARRANGE	$\Delta\theta = \dfrac{\Delta E}{m\,c}$
SUBSTITUTE	$\Delta\theta = \dfrac{7800}{0.50 \times 390}$
ANSWER	$\Delta\theta = 40$ °C

YOUR TURN

27 000 J raises the temperature of a 1.5 kg metal block by 20 °C. Calculate the specific heat capacity of the metal.

KNOW	**$\Delta E = 27\,000$ J, m = 1.5 kg, $\Delta\theta = 20$ °C**
REARRANGE	$c = \dfrac{\Delta E}{m\,\Delta\theta}$
SUBSTITUTE	$c = \dfrac{27\,000}{1.5 \times 20}$
ANSWER	**c = 900 J/kg °C**

Get the marks: $\Delta\theta$ is the **change** in temperature, not the final temperature. From 20 °C to 100 °C is a change of 80 °C. If the question asks for the final temperature, add your $\Delta\theta$ to the starting temperature.

HEAT CAPACITY OR LATENT HEAT?

	SPECIFIC HEAT CAPACITY	SPECIFIC LATENT HEAT
What happens	the temperature changes	the state changes
Temperature	changes	stays constant
Equation	$\Delta E = m\,c\,\Delta\theta$	$E = m\,L$
Unit	J/kg °C	J/kg

PRACTISE

TRY IT

1 Draw a ring around the correct answer in each box.

The unit of specific heat capacity is **J/kg** | **J/kg °C** | **J/°C**

The same energy is given to 1 kg of water and to 1 kg of copper. The bigger temperature rise is in the

water | **copper**

2 **SUPPORTED** Calculate the energy needed to raise the temperature of 3.0 kg of iron by 50 °C. Specific heat capacity of iron = 450 J/kg °C

EQUATION	$\Delta E = m\,c\,\Delta\theta$
SUBSTITUTE	$\Delta E = 3.0 \times 450 \times 50$
ANSWER	**67 500 J** unit:

3 **SUPPORTED** 250 g of water is heated from 15 °C to 95 °C. Calculate the energy transferred to the water. Specific heat capacity of water = 4200 J/kg °C

CONVERT	**m = 0.250 kg**
TEMPERATURE CHANGE	$\Delta\theta = 95 - 15 = 80$ °C
EQUATION	$\Delta E = m\,c\,\Delta\theta$
SUBSTITUTE	$\Delta E = 0.250 \times 4200 \times 80$
ANSWER	**84 000 J** unit:

4 Complete the table. Show your working underneath.

Substance	Mass / kg	c / J/kg °C	$\Delta\theta$ / °C	ΔE / J
Cooking oil	0.50	2000	60	60 000
Lead	2.0	130	20	5200
Unknown metal	4.0	450	10	18 000

5 For each change, write the equation you would use.

Change	Equation
Heating water from 20 °C to 80 °C	$\Delta E = m\,c\,\Delta\theta$
Melting ice at 0 °C	$E = m\,L$
Boiling water at 100 °C	$E = m\,L$
Cooling a hot iron bar	$\Delta E = m\,c\,\Delta\theta$

6 CHALLENGE A kettle heats 0.20 kg of water from 20 °C to 100 °C, and then boils all of it away. Calculate the total energy transferred to the water. Use c = 4200 J/kg °C and L = 2 260 000 J/kg

Heating: $\Delta E = m\,c\,\Delta\theta = 0.20 \times 4200 \times 80 = 67\,200$ J. Boiling:

$E = m\,L = 0.20 \times 2\,260\,000 = 452\,000$ J. Total = 519 200 J.

AQA-STYLE

EXAM CORNER

Write in full sentences. Show every step of a calculation.

Get the marks: Work out $\Delta\theta$ first and write it down. It is the step most often missed.

7 A hot-water bottle holds 0.80 kg of water. The water cools from 70 °C to 30 °C. Calculate the energy transferred from the water. Specific heat capacity of water = 4200 J/kg °C **[3 marks]**

TEMPERATURE CHANGE	$\Delta\theta = 70 - 30 = 40$ °C (1)
EQUATION	$\Delta E = m\,c\,\Delta\theta$
SUBSTITUTE	$\Delta E = 0.80 \times 4200 \times 40$ (1)
ANSWER	**134 400 J (1)** unit:

8 A heater transfers 36 000 J to a 2.0 kg block of aluminium. Calculate the temperature rise of the block. Specific heat capacity of aluminium = 900 J/kg °C **[3 marks]**

EQUATION	$\Delta E = m\,c\,\Delta\theta$
REARRANGE	$\Delta\theta = \dfrac{\Delta E}{m\,c}$ (1)
SUBSTITUTE	$\Delta\theta = \dfrac{36\,000}{2.0 \times 900}$ (1)
ANSWER	**20 °C (1)** unit:

9 Give one difference between specific heat capacity and specific latent heat. **[2 marks]**

Specific heat capacity is about a change in temperature (energy to raise 1 kg by 1 °C) (1); specific latent heat is about a change of state at constant temperature (energy to change the state of 1 kg) (1).

REVIEW LATER

REVISE IT

Use this box after the lesson and before tests.

SUMMARY

- A temperature rise depends on the energy supplied, the mass and the material.
- Specific heat capacity c: the energy to raise the temperature of 1 kg by 1 °C, in J/kg °C.
- $\Delta E = m\,c\,\Delta\theta$; rearranged, $c = \dfrac{\Delta E}{m\,\Delta\theta}$ and $\Delta\theta = \dfrac{\Delta E}{m\,c}$.
- Temperature changing: specific heat capacity. State changing: specific latent heat.

COVER THE PAGE AND ANSWER

1 Define specific heat capacity.

2 Write the equation for a change in thermal energy.

3 State the unit of specific heat capacity.

4 What does $\Delta\theta$ stand for?

I CAN…

- state what a temperature rise depends on
- define specific heat capacity
- calculate with $\Delta E = m\,c\,\Delta\theta$
- rearrange to find c or $\Delta\theta$
- choose between specific heat capacity and specific latent heat

REFERENCE

EQUATION TOOLKIT

Every equation from Lessons 1–7. AQA prints all of them on the equation sheet in the exam, but you still need to know what each symbol means, its unit, and how to use it.

1. Cover the quantity you want to find.
2. What is left side by side? **Multiply**.
3. One above the other? **Divide** the top by the bottom.

Example: to find V from $\rho = \frac{m}{V}$, cover V. m is above ρ, so $V = \frac{m}{\rho}$.

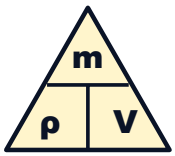
$\rho = \frac{m}{V}$

Lesson 3

density = mass ÷ volume

kg/m³, kg, m³ (or g/cm³, g, cm³)

$m = \rho V$ and $V = \frac{m}{\rho}$



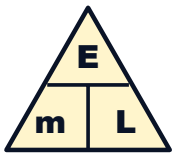
$E = m L$

Lesson 6

energy for a change of state = mass × specific latent heat

J, kg, J/kg

$L = \frac{E}{m}$ and $m = \frac{E}{L}$



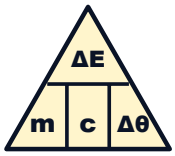
$\Delta E = m c \Delta\theta$

Lesson 7

change in thermal energy = mass × specific heat capacity × temperature change

J, kg, J/kg °C, °C

$c = \frac{\Delta E}{m \Delta\theta}$ and $\Delta\theta = \frac{\Delta E}{m c}$



1

Write what you know. Convert first: g → kg, cm³ → m³, kJ → J.

2

Write the equation in symbols.

3

Rearrange if you need to, then substitute the numbers.

4

Give the answer with a unit. Ask: is it sensible?

REFERENCE

KEY WORDS FROM THIS LESSON

Learn the meaning and the spelling. In exams, precise words earn marks: say “the particles vibrate about fixed positions”, not “the particles shake”.

Specific heat capacity (c): The energy needed to raise the temperature of 1 kg of a substance by 1 °C, in J/kg °C

Temperature change (Δθ): Final temperature minus starting temperature, in °C

CHECK YOURSELF

QUICK ANSWERS

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in the answer edition.

DO NOW

1. $E = m L$
2. Kinetic energy.
3. 63 − 18 = 45 °C

COVER THE PAGE

1. The energy needed to raise the temperature of 1 kg of a substance by 1 °C.
2. $\Delta E = m c \Delta \theta$
3. J/kg °C
4. The change in temperature.